

EDA_Titanic_Task5

September 8, 2026

```
[2]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset directly from a live web link
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/
↪titanic.csv"
df = pd.read_csv(url)

# Save a local copy to your folder automatically
df.to_csv('titanic.csv', index=False)

# Preview the first 5 rows
df.head()
```

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[2]:
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	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

```
[3]: # Check data types and missing values
print("--- DATA INFO ---")
df.info()

print("\n--- STATISTICAL SUMMARY ---")
display(df.describe())

print("\n--- CATEGORICAL VALUE COUNTS ---")
print("Survived:\n", df['Survived'].value_counts())
print("\nSex:\n", df['Sex'].value_counts())
print("\nPclass:\n", df['Pclass'].value_counts())
```

```
--- DATA INFO ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null   int64
1   Survived         891 non-null   int64
2   Pclass           891 non-null   int64
3   Name             891 non-null   object
4   Sex              891 non-null   object
5   Age              714 non-null   float64
6   SibSp            891 non-null   int64
7   Parch            891 non-null   int64
8   Ticket           891 non-null   object
9   Fare             891 non-null   float64
10  Cabin            204 non-null   object
11  Embarked         889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

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--- STATISTICAL SUMMARY ---
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	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208

```
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

--- CATEGORICAL VALUE COUNTS ---

Survived:

```
0      549
1      342
```

Name: Survived, dtype: int64

Sex:

```
male      577
female    314
```

Name: Sex, dtype: int64

Pclass:

```
3      491
1      216
2      184
```

Name: Pclass, dtype: int64

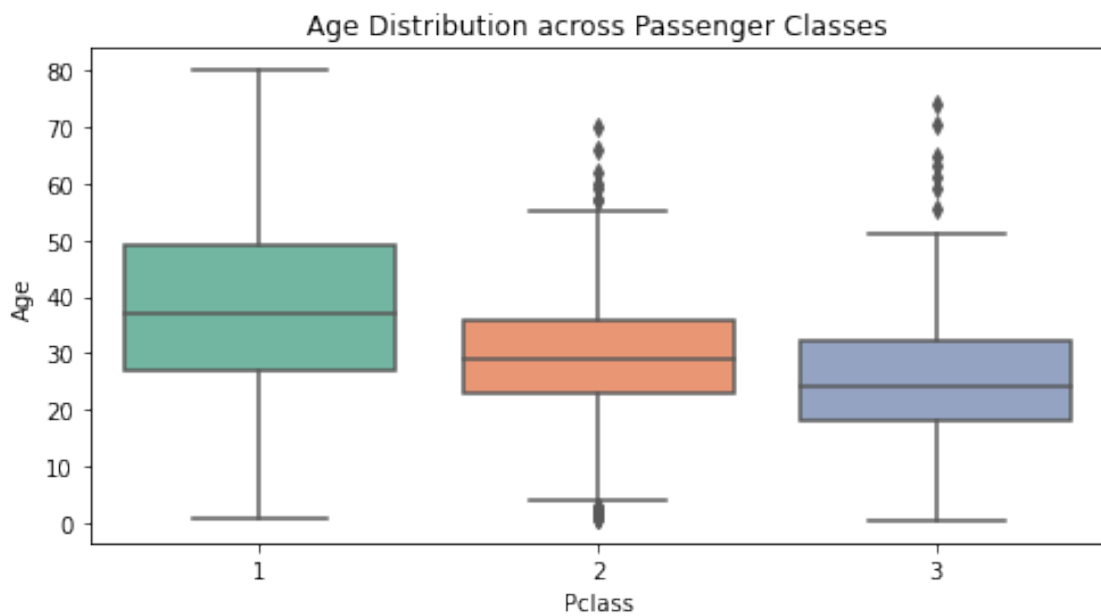
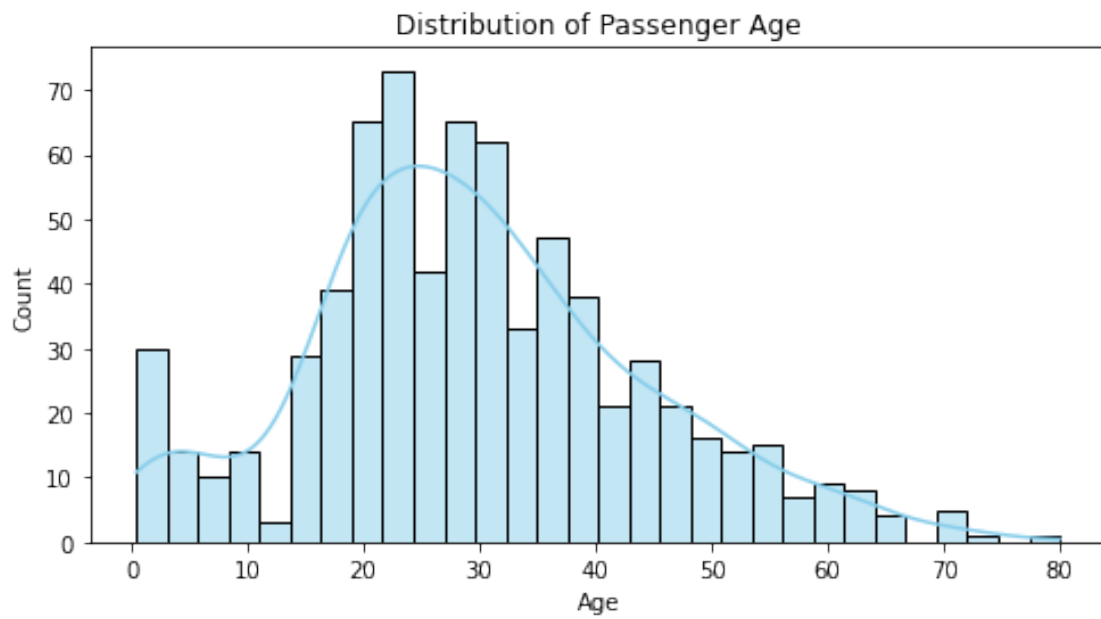
```
[4]: # 1. Distribution Plot (Histogram) for Age
plt.figure(figsize=(8, 4))
sns.histplot(df['Age'].dropna(), kde=True, bins=30, color='skyblue')
plt.title('Distribution of Passenger Age')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()

# 2. Boxplot (Age by Passenger Class)
plt.figure(figsize=(8, 4))
sns.boxplot(x='Pclass', y='Age', data=df, palette='Set2')
plt.title('Age Distribution across Passenger Classes')
plt.show()

# 3. Correlation Heatmap
plt.figure(figsize=(8, 5))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Feature Correlation Heatmap')
plt.show()

# 4. Pairplot for Key Features
sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']].dropna(),
             hue='Survived', palette='bwr')
```

```
plt.show()
```



TypeError

Input In [4], in <cell line: 17>()

15 # 3. Correlation Heatmap

Traceback (most recent call last)

```

16 plt.figure(figsize=(8, 5))
----> 17 sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm',
    ↪fmt=".2f")
18 plt.title('Feature Correlation Heatmap')
19 plt.show()

```

`TypeError: corr() got an unexpected keyword argument 'numeric_only'`

<Figure size 576x360 with 0 Axes>

```

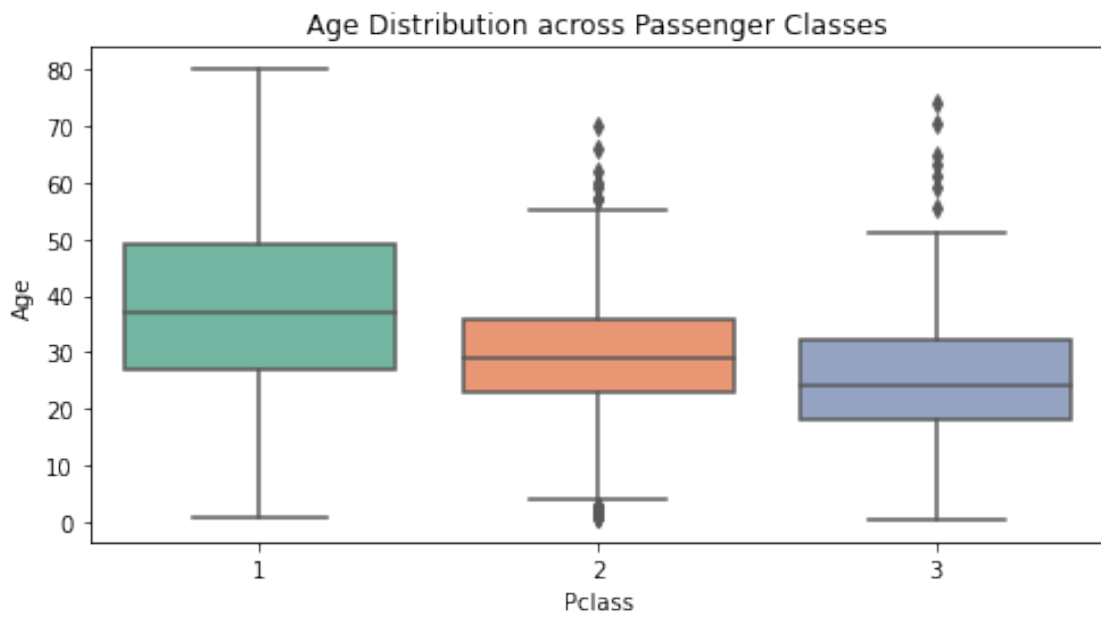
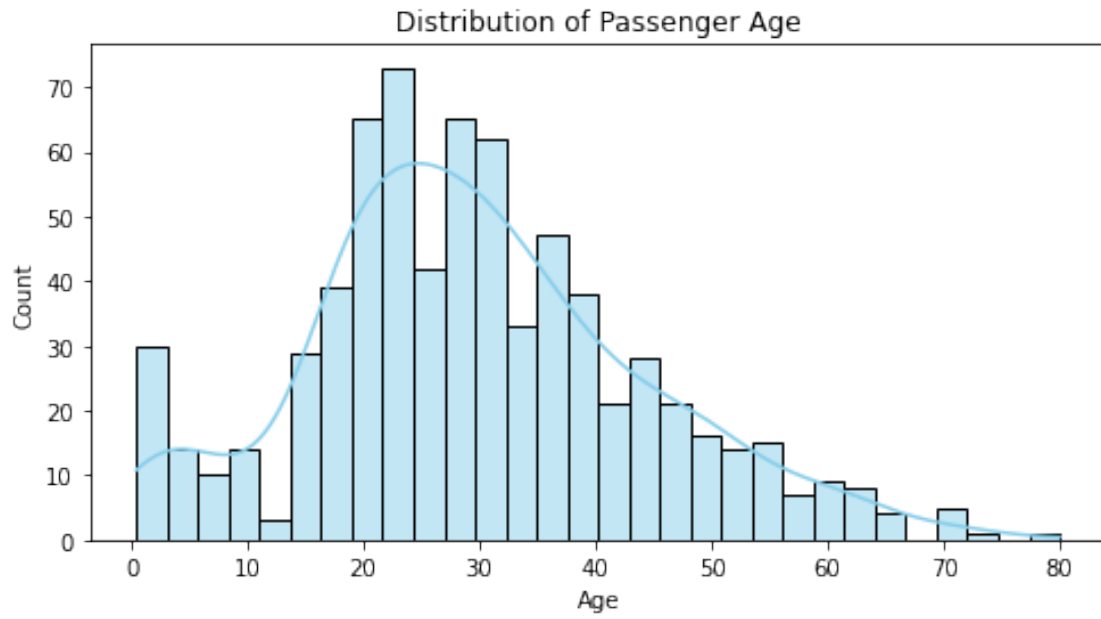
[5]: # 1. Distribution Plot (Histogram) for Age
plt.figure(figsize=(8, 4))
sns.histplot(df['Age'].dropna(), kde=True, bins=30, color='skyblue')
plt.title('Distribution of Passenger Age')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()

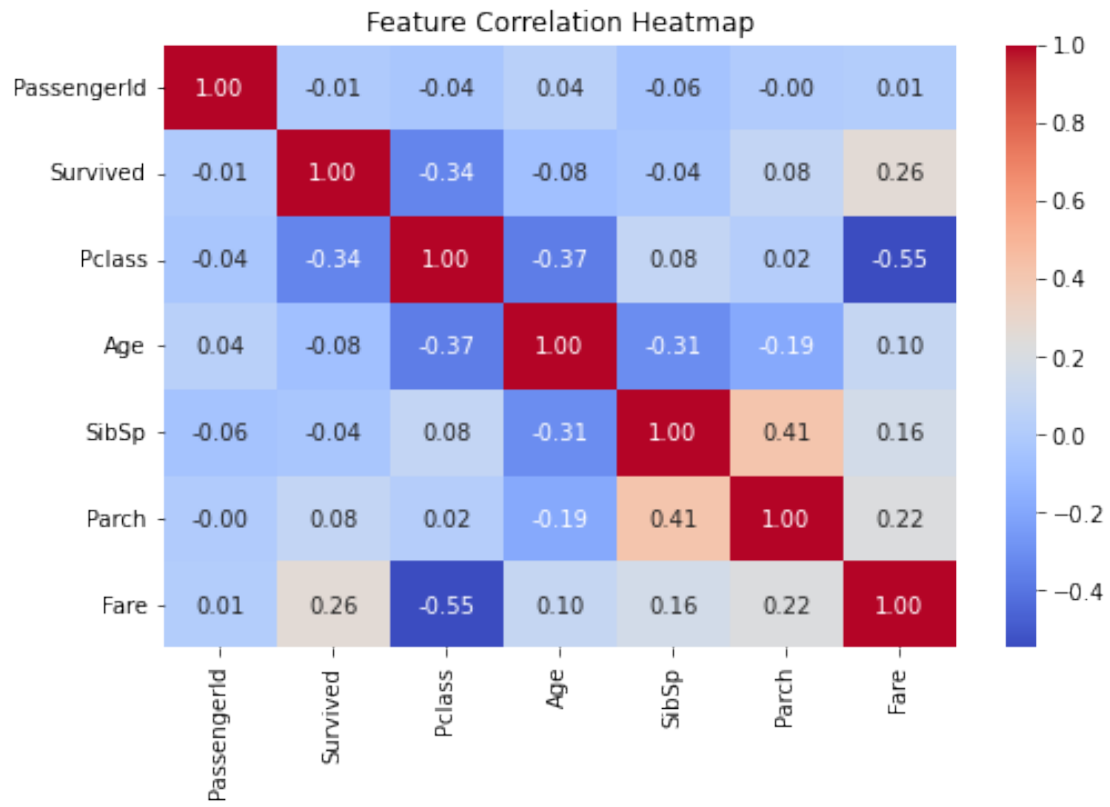
# 2. Boxplot (Age by Passenger Class)
plt.figure(figsize=(8, 4))
sns.boxplot(x='Pclass', y='Age', data=df, palette='Set2')
plt.title('Age Distribution across Passenger Classes')
plt.show()

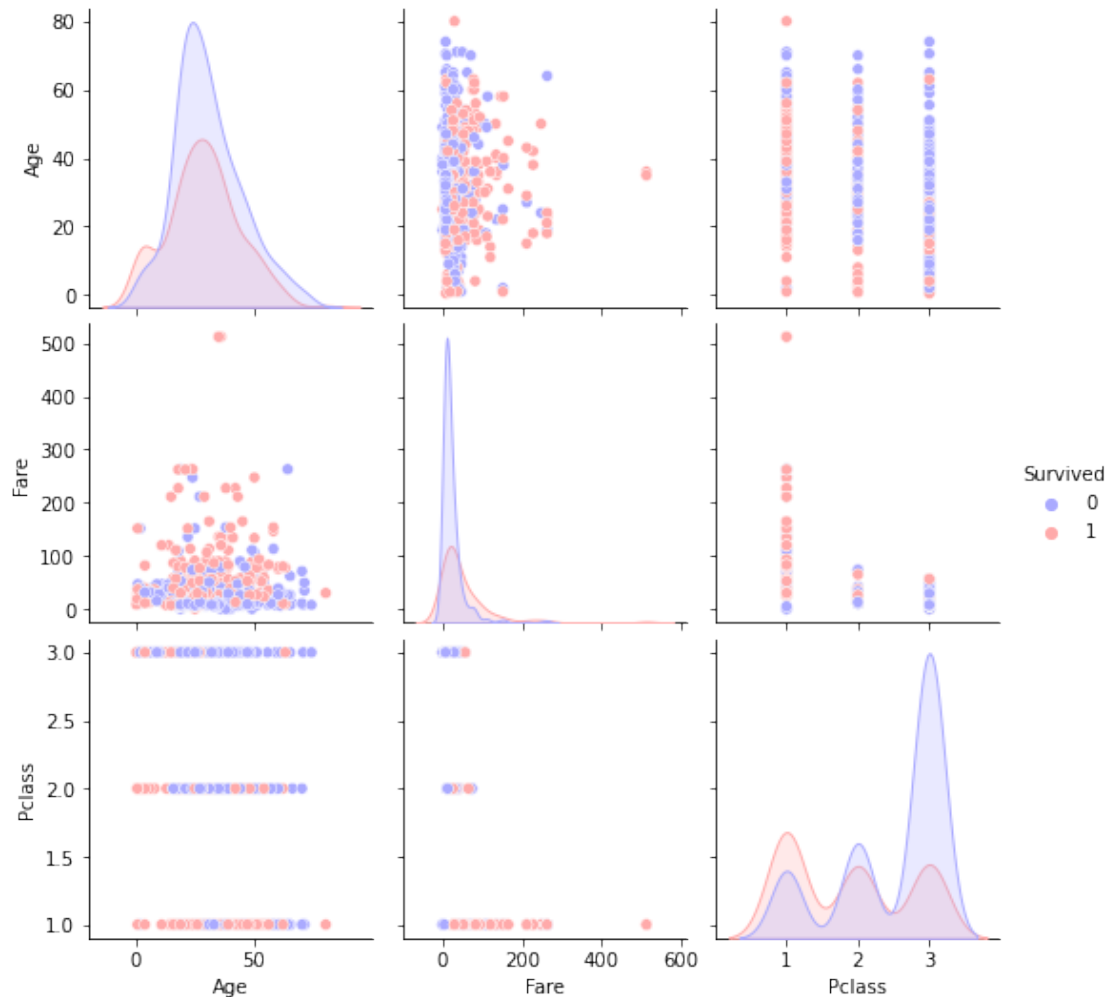
# 3. Correlation Heatmap (Fixed for older Pandas versions)
plt.figure(figsize=(8, 5))
numeric_df = df.select_dtypes(include=['number'])
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Feature Correlation Heatmap')
plt.show()

# 4. Pairplot for Key Features
sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']].dropna(),
    ↪hue='Survived', palette='bwr')
plt.show()

```







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0.1 Key Observations & Insights

- **Age Distribution:** The passenger age distribution is right-skewed, peaking between ages 20 and 30, with a small spike in infants.
- **Class vs. Age:** First-class passengers skew older (median age around 38), while third-class passengers are younger (median age around 25).
- **Correlation:** Ticket class (Pclass) shows a negative correlation with survival rate, showing higher-class passengers were prioritized.
- **Summary of Findings:** Survival rates favored women, children, and passengers holding 1st class tickets.

0.2 Interview Questions & Answers

1. **What is EDA and why is it important?** EDA (Exploratory Data Analysis) is the process of analyzing datasets using visual and statistical tools. It helps identify patterns, spot anomalies, and test assumptions before modeling.
2. **Which plots do you use to check correlation?** Heatmaps (`sns.heatmap()`) and scatterplots/pairplots (`sns.pairplot()`).
3. **How do you handle skewed data?** By using transformations (such as log, square root, or Box-Cox) or capping/removing extreme outliers.
4. **How to detect multicollinearity?** By calculating the Variance Inflation Factor (VIF) or inspecting correlation heatmaps for values near +1 or -1.
5. **What are univariate, bivariate, and multivariate analyses?**
 - **Univariate:** Analyzing one variable (e.g., Age distribution histogram).
 - **Bivariate:** Analyzing two variables (e.g., Boxplot of Age vs. Pclass).
 - **Multivariate:** Analyzing three or more variables (e.g., Pairplot or Heatmap).
6. **Difference between heatmap and pairplot?** A heatmap shows numerical correlation scores in a color-coded matrix, while a pairplot displays pairwise scatter plots and distributions.
7. **How do you summarize your insights?** By presenting concise bullet points combining key statistical metrics, trends, and actionable takeaways.

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